

## LAKE COUNTY AP, OR, USA

WMO: 725976

Lat: 42.167N

Lon: 120.400W

Elev: 1441

StdP: 85.17

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94285

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -16.4      | -12.8 | -22.1                         | 0.6 | -12.3 | -17.4 | 1.0 | -12.1 | 12.3                  | 3.5  | 11.1 | 1.8  | 1.8                   | 350  | 0.647 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.0                   | 32.5            | 15.1 | 31.1 | 14.5 | 29.1 | 13.8 | 16.3                | 29.5 | 15.4 | 28.1 | 14.7 | 27.1 | 3.6                  | 190  |

| Dehumidification DP/MCDB and HR |      |      |      |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 11.9                            | 10.3 | 20.1 | 10.1 | 9.2 | 19.2 | 8.8 | 8.4 | 18.8 | 50.7          | 29.3 | 48.0 | 28.0 | 45.7 | 27.6 | 20.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 10.5          | 9.0  | 8.1 | DB                         | -21.2 | 35.1               | 3.0 | 2.0                                                | -23.4 | 36.5       | -25.2 | 37.7       | -26.9 | 38.8       | -29.0 | 40.2 | (4) |
| (5)               |      |     | WB                         | -21.6 | 17.8               | 4.0 | 1.0                                                | -24.5 | 18.6       | -26.9 | 19.2       | -29.1 | 19.8       | -32.1 | 20.5 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 7.5         | -1.5  | -0.1  | 3.0   | 5.7   | 10.2  | 14.6  | 19.3  | 17.8  | 13.8  | 7.3   | 1.9   | -2.3  | (6)  |  |
|                                                                        | DBStd       | 8.29        | 4.53  | 4.15  | 3.83  | 3.86  | 4.09  | 4.09  | 3.21  | 3.03  | 3.83  | 3.75  | 4.43  | 4.73  | (7)  |  |
|                                                                        | HDD10.0     | 1789        | 357   | 282   | 217   | 138   | 48    | 9     | 0     | 1     | 12    | 99    | 244   | 381   | (8)  |  |
|                                                                        | HDD18.3     | 4051        | 616   | 516   | 475   | 380   | 252   | 124   | 26    | 46    | 142   | 342   | 494   | 639   | (9)  |  |
|                                                                        | CDD10.0     | 881         | 0     | 0     | 1     | 8     | 56    | 146   | 288   | 242   | 125   | 16    | 1     | 0     | (10) |  |
|                                                                        | CDD18.3     | 101         | 0     | 0     | 0     | 0     | 1     | 11    | 56    | 29    | 4     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 3037        | 0     | 0     | 0     | 2     | 63    | 337   | 1275  | 961   | 376   | 23    | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 1110        | 0     | 0     | 0     | 0     | 8     | 93    | 548   | 357   | 99    | 4     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 3.6         | 3.3   | 3.6   | 3.9   | 4.0   | 3.8   | 3.7   | 3.4   | 3.4   | 3.3   | 3.5   | 3.4   | 3.5   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 400         | 52    | 39    | 38    | 33    | 38    | 28    | 9     | 11    | 16    | 27    | 51    | 57    | (15) |  |
|                                                                        | PrecMax     | 613         | 148   | 120   | 98    | 86    | 111   | 107   | 35    | 77    | 69    | 168   | 117   | 228   | (16) |  |
|                                                                        | PrecMin     | 248         | 5     | 3     | 7     | 5     | 4     | 0     | 0     | 0     | 0     | 5     | 0     | 0     | (17) |  |
|                                                                        | PrecStd     | 95          | 35    | 25    | 21    | 19    | 26    | 26    | 10    | 16    | 17    | 28    | 32    | 43    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 11.2  | 14.6  | 18.8  | 22.9  | 27.6  | 32.4  | 35.3  | 33.7  | 31.3  | 27.1  | 18.1  | 9.2   | (19) |  |
|                                                                        |             | MCWB        | 5.5   | 6.8   | 8.1   | 10.4  | 13.3  | 15.6  | 16.5  | 15.2  | 13.6  | 11.8  | 8.5   | 5.0   | (20) |  |
|                                                                        | 2%          | DB          | 7.9   | 11.3  | 16.1  | 20.9  | 25.0  | 29.0  | 33.0  | 32.1  | 29.0  | 22.5  | 15.8  | 7.6   | (21) |  |
|                                                                        |             | MCWB        | 4.1   | 5.1   | 7.2   | 10.0  | 12.4  | 14.2  | 15.5  | 14.6  | 12.8  | 10.2  | 7.6   | 4.1   | (22) |  |
|                                                                        | 5%          | DB          | 6.3   | 8.0   | 13.0  | 17.6  | 22.7  | 27.3  | 32.0  | 30.7  | 27.5  | 20.2  | 12.2  | 6.1   | (23) |  |
|                                                                        |             | MCWB        | 3.0   | 3.7   | 6.2   | 8.3   | 11.5  | 13.4  | 15.0  | 14.1  | 12.3  | 9.3   | 6.1   | 3.2   | (24) |  |
|                                                                        | 10%         | DB          | 4.0   | 6.2   | 10.9  | 14.3  | 20.2  | 25.0  | 30.0  | 28.7  | 25.9  | 17.4  | 9.0   | 4.0   | (25) |  |
|                                                                        |             | MCWB        | 1.7   | 2.6   | 5.6   | 6.7   | 10.4  | 12.6  | 14.5  | 13.5  | 11.8  | 8.0   | 4.9   | 1.8   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 6.8   | 7.6   | 9.2   | 11.7  | 14.8  | 16.7  | 18.0  | 16.6  | 15.2  | 12.3  | 9.8   | 6.4   | (27) |  |
|                                                                        |             | MCDB        | 9.2   | 13.4  | 16.8  | 20.3  | 24.8  | 29.2  | 30.8  | 30.2  | 25.4  | 24.3  | 17.1  | 8.2   | (28) |  |
|                                                                        | 2%          | WB          | 4.8   | 6.0   | 7.9   | 10.1  | 13.4  | 15.3  | 16.7  | 15.4  | 13.9  | 11.0  | 7.9   | 4.7   | (29) |  |
|                                                                        |             | MCDB        | 7.7   | 10.5  | 14.4  | 19.4  | 23.3  | 26.8  | 30.1  | 28.7  | 26.2  | 21.5  | 13.9  | 6.9   | (30) |  |
|                                                                        | 5%          | WB          | 3.4   | 4.4   | 6.7   | 8.8   | 12.2  | 14.1  | 15.8  | 14.7  | 12.9  | 9.8   | 6.7   | 3.5   | (31) |  |
|                                                                        |             | MCDB        | 5.6   | 7.7   | 12.3  | 16.8  | 21.2  | 25.2  | 29.0  | 27.8  | 25.7  | 18.8  | 11.6  | 5.5   | (32) |  |
|                                                                        | 10%         | WB          | 2.2   | 3.0   | 5.6   | 7.5   | 10.9  | 13.1  | 15.1  | 14.0  | 12.1  | 8.6   | 5.3   | 2.4   | (33) |  |
|                                                                        |             | MCDB        | 4.1   | 5.6   | 10.4  | 14.1  | 19.1  | 23.7  | 27.9  | 27.0  | 24.9  | 16.6  | 9.0   | 4.1   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.6   | 10.5  | 11.7  | 13.2  | 14.9  | 16.9  | 20.0  | 20.7  | 20.0  | 17.3  | 12.5  | 10.0  | (35) |  |
|                                                                        |             | MCDBR       | 12.3  | 14.1  | 17.3  | 19.6  | 19.9  | 20.3  | 21.6  | 22.8  | 23.7  | 23.3  | 19.0  | 12.0  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 8.7   | 9.4   | 10.4  | 10.9  | 10.0  | 9.2   | 8.8   | 9.7   | 11.2  | 12.7  | 12.0  | 8.7   | (37) |  |
|                                                                        |             | MCWBR       | 10.7  | 12.7  | 15.6  | 18.7  | 18.4  | 19.1  | 20.3  | 20.7  | 22.2  | 21.2  | 16.7  | 9.1   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.264 | 0.274 | 0.274 | 0.302 | 0.305 | 0.296 | 0.306 | 0.303 | 0.292 | 0.289 | 0.267 | 0.264 | (40) |  |
|                                                                        |             | taud        | 2.546 | 2.558 | 2.597 | 2.508 | 2.541 | 2.596 | 2.567 | 2.557 | 2.574 | 2.574 | 2.602 | 2.547 | (41) |  |
|                                                                        | Ebn at Noon |             | 906   | 949   | 986   | 970   | 969   | 973   | 960   | 957   | 950   | 916   | 897   | 873   | (42) |  |
|                                                                        |             | Edh at Noon | 68    | 79    | 86    | 101   | 101   | 96    | 98    | 95    | 87    | 77    | 64    | 63    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.89  | 2.79  | 4.00  | 5.40  | 6.61  | 7.74  | 7.93  | 6.94  | 5.49  | 3.64  | 2.15  | 1.51  | (44) |  |
|                                                                        | RadStd      |             | 0.26  | 0.34  | 0.37  | 0.48  | 0.58  | 0.41  | 0.33  | 0.35  | 0.28  | 0.28  | 0.22  | 0.27  | (45) |  |

## Historical Trends

|                             |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |
| (47) Regional (7 neighbors) |  | N/A   | N/A     | -1.21  | +0.62 | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |

Nomenclature: See separate page